

**LAB REPORT**  
**FOR**  
**LAB-5: FRAME DETECTION AND FREQUENCY OFFSET**  
**CORRECTION**

**EEC 556: SOFTWARE DEFINED RADIO**

**DR. MURAD HIZLAN**

Associate Professor and chair  
Electrical & Computer Engineering

**NIKHIL REDDY KUNTLOOR**

**2909957**



**04/28/2025**

## INTRODUCTION:

In this lab, we are using the training sequence for frame synchronization and frequency offset correction. In reality, both of these operations are synchronization operations. The reason for frame synchronization is so that the receiver will know where each frame begins, and as such, where the data symbols begin (right after the training sequence). Therefore, the receiver is basically looking for a “match” to the training sequence in the received sequence of the symbols. When the match is detected, the receiver knows that it has identified the training sequence (also known as the reference sequence in this context), and so the data symbols will follow.

In this lab we have designed both these virtual instruments(VIs):

1. **Moose.vi**
2. **Sliding correlator.vi**

### 1. Moose.vi

We have downloaded the lab files and modified with our own version of it, moose.vi is used to estimate and correct the frequency offset while implementing formula as shown in figure 1.

$$\epsilon = \frac{\text{phase} \sum_{l=L}^{N_t-1} y[l + N_t] y^*[l]}{2\pi N_t}$$

$$\tilde{y}[n] = e^{-j2\pi\epsilon n} y[n]$$

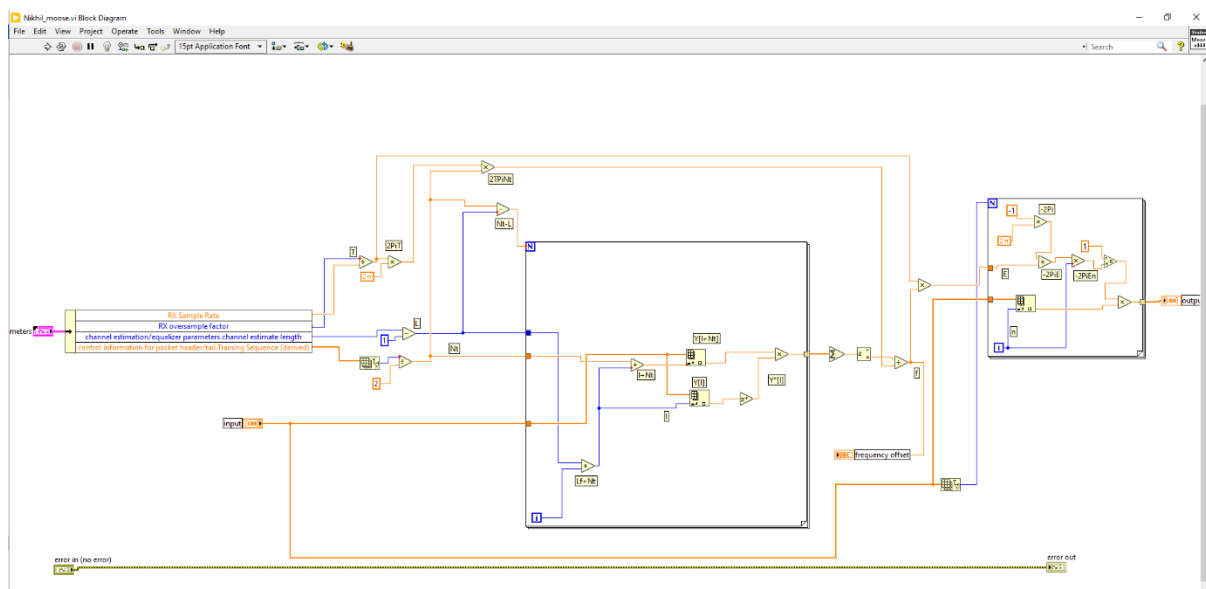


Figure 1: Nikhil\_moose.vi(block diagram)

### 2. Sliding correlator.vi

Sliding correlator.vi is used for frame detection (synchronization), in this we need to call Moose.vi with the proper inputs wired to get an estimate of the frequency offset if the input is true, we used the frequency offset corrected y output from Moose.vi to feed to the

sliding\_correlator.vi y output. Otherwise, we will just bypass Moose.vi and feed the frame offset corrected y directly to the y output. As shown in figure 2.

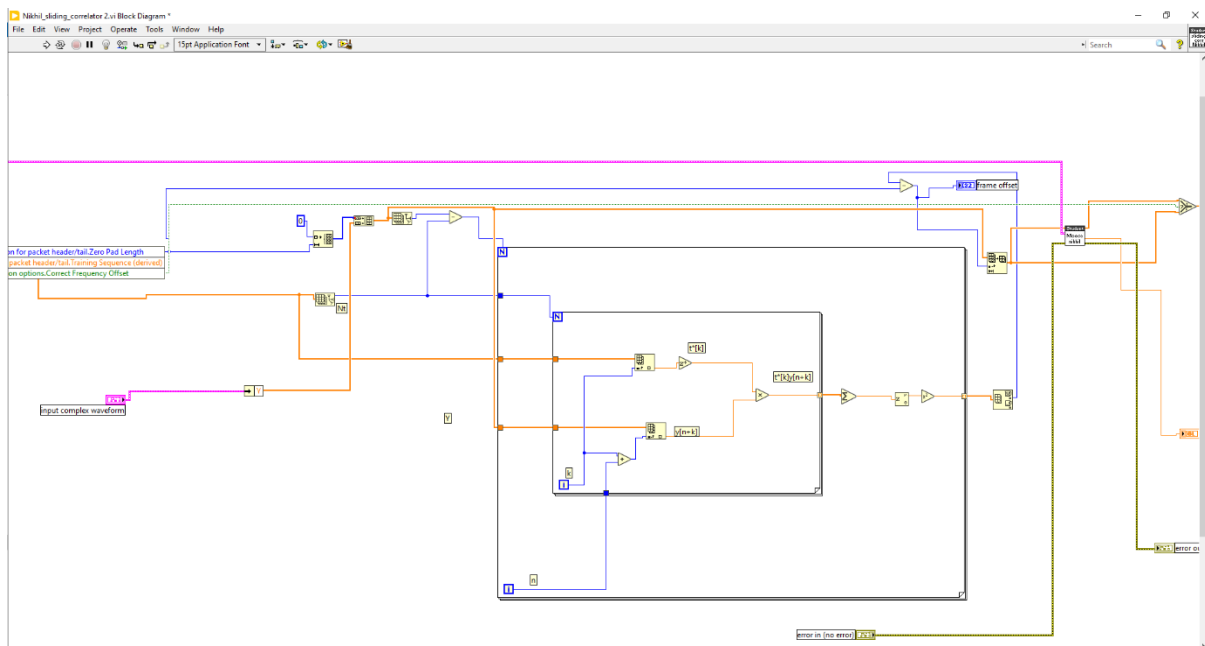


Figure 2: Nikhil\_sliding\_correlator.vi(block diagram)

Receiver.vi opening up synchronize.vi(double click), selecting frame detection case and opening up frame\_detect.vi, selecting the sliding correlator case and replaced sliding\_correlator.vi with our implementation as shown in figure 3.

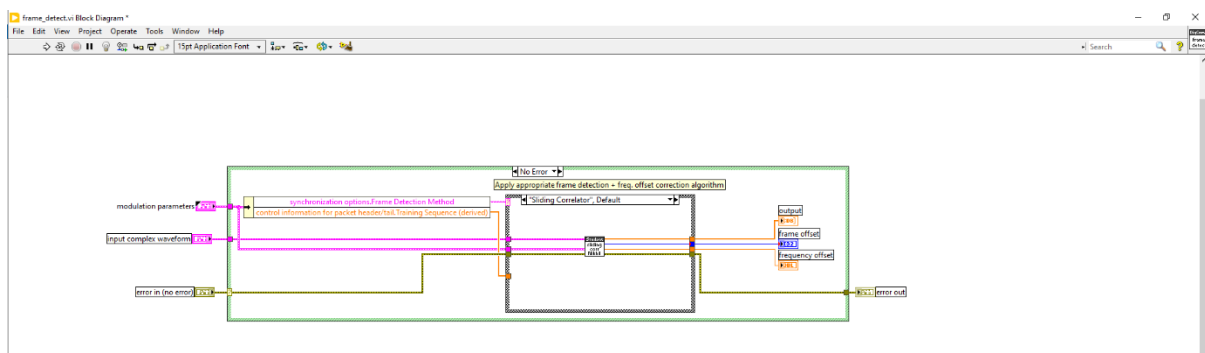


Figure 3: frame\_detect.vi (block diagram)

We have observed the results in simulator.vi as shown in figure 4,5,6,7.

On the simulator.vi we have set following parameters as shown below

- TX oversample factor = RX oversample factor = 10
- TX sample rate = RX sample rate = 10MHz
- Packet length(bits) = 500
- Synchronization method = Timing estimation
- ISI Channel =  $\{h[0] = 0.35e^{j\pi/4} = 0.25 + 0.25i\}$

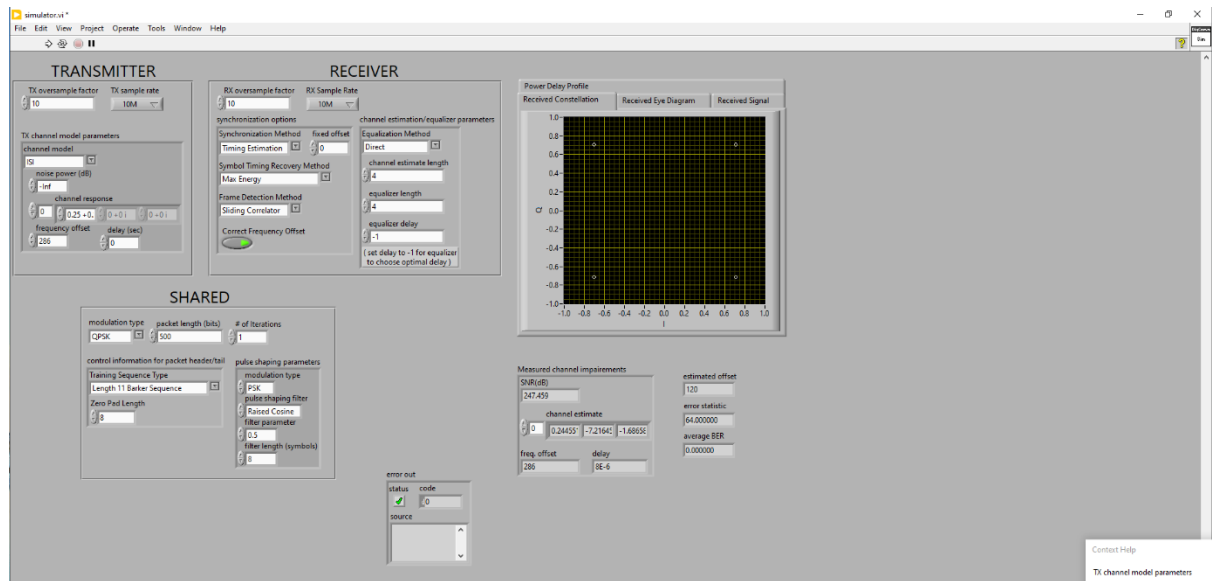


Figure 4: simulator.vi(received constellation)

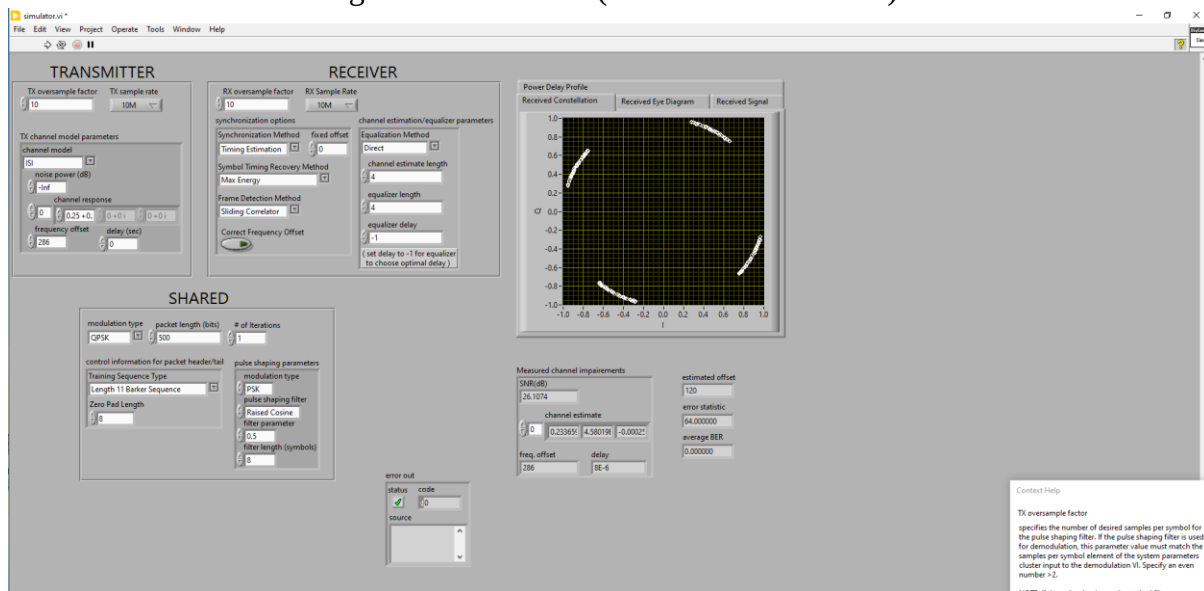


Figure 5: simulator.vi(received constellation)

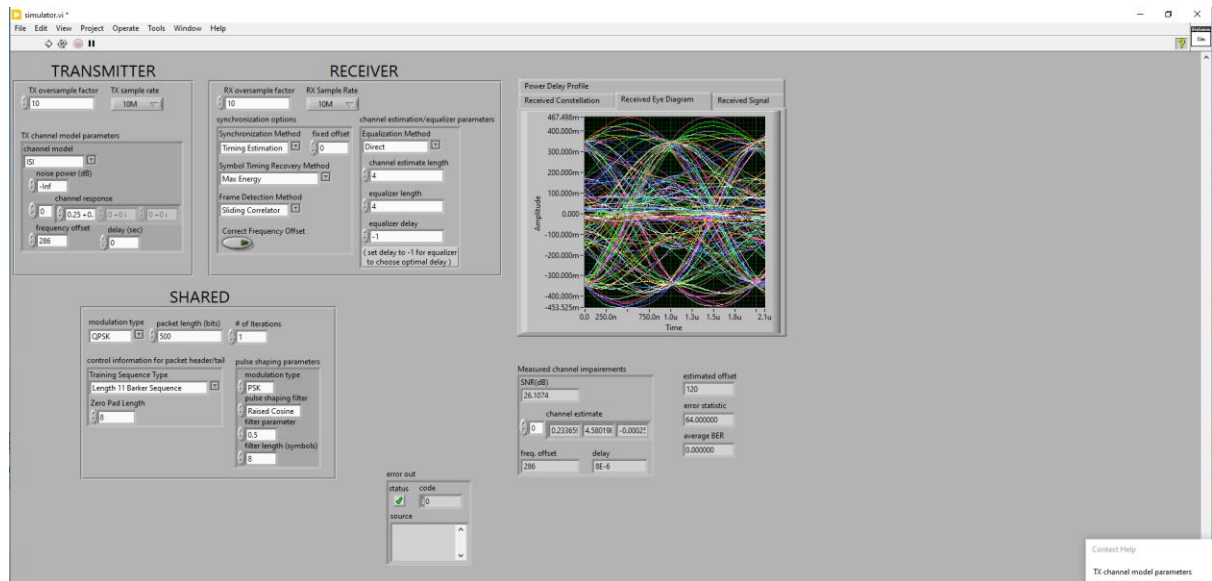


Figure 6: simulator.vi(Received eye diagram)

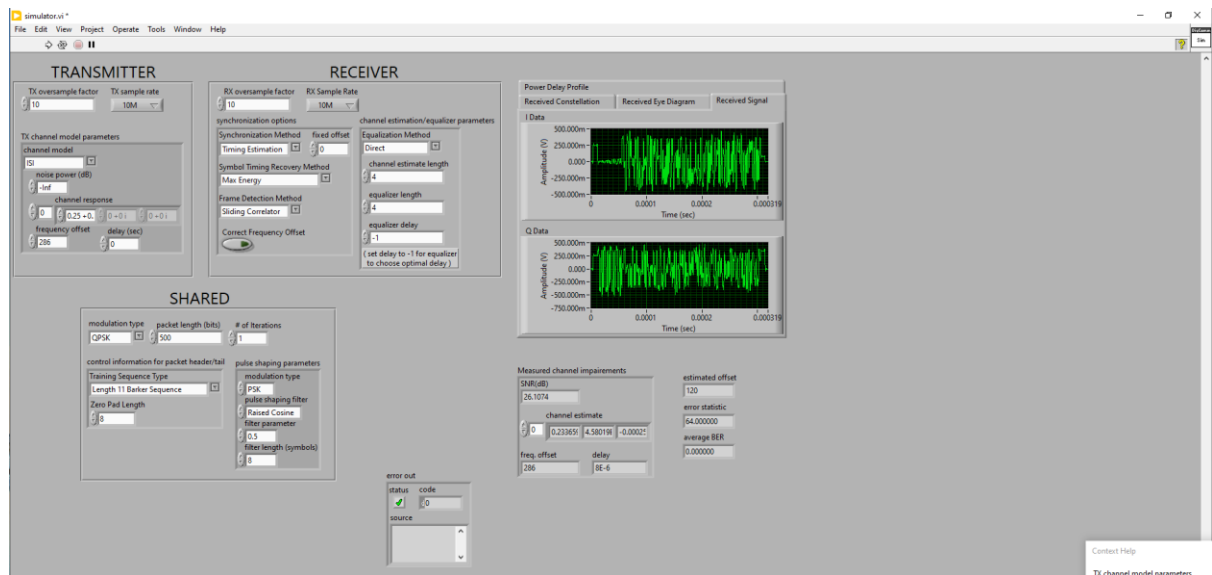


Figure 7: simulator.vi(Received signal)

**Question and answers:**

**2. Set up your simulator according to the steps used for verifying your implementation of the Moose algorithm. Set the Correct Frequency Offset control to FALSE. Describe what happens to the received constellation if you do not correct for the 201 Hz frequency offset.**

**Ans:** If we set the correct frequency offset control by setting it to FALSE, the received constellation will experience a phase shift, causing it to appear shifted or rotated from its original position.

**3. Based on the system parameters described in the pre-lab, what are the range of frequency offsets that you can estimate/correct using the Moose algorithm? Since the Moose algorithm cannot estimate/correct for arbitrary frequency offsets it is often considered to perform what is known as fine frequency synchronization.**

**Ans:** According to the parameter set in simulator,

TX oversample factor = RX oversample factor = 10.

TX sample rate = RX sample rate = 10 MHz.

So Symbol Rate = TX oversample factor / TX sample rate = 1 MHz .

Maximum frequency offset correction by “Moose” algorithm Given by = Symbol Rate / 2.

= 1MHz/2 = 500kHz.

## LAB EXPERIMENT:

We have connected NI USRP devices to computer and run top\_tx.vi and top\_rx.vi to observe the results as shown in figure 8, 9, 10.

We have set the following parameters on top\_tx.vi and top\_rx.vi in received QAM constellation

- TX sample rate = 4M samp/sec
- TX oversample factor = 20
- RX sample rate = 4M samp/sec
- RX oversample factor = 20
- Channel estimate length = 2
- Equalizer length = 4
- Capture time = 2msec

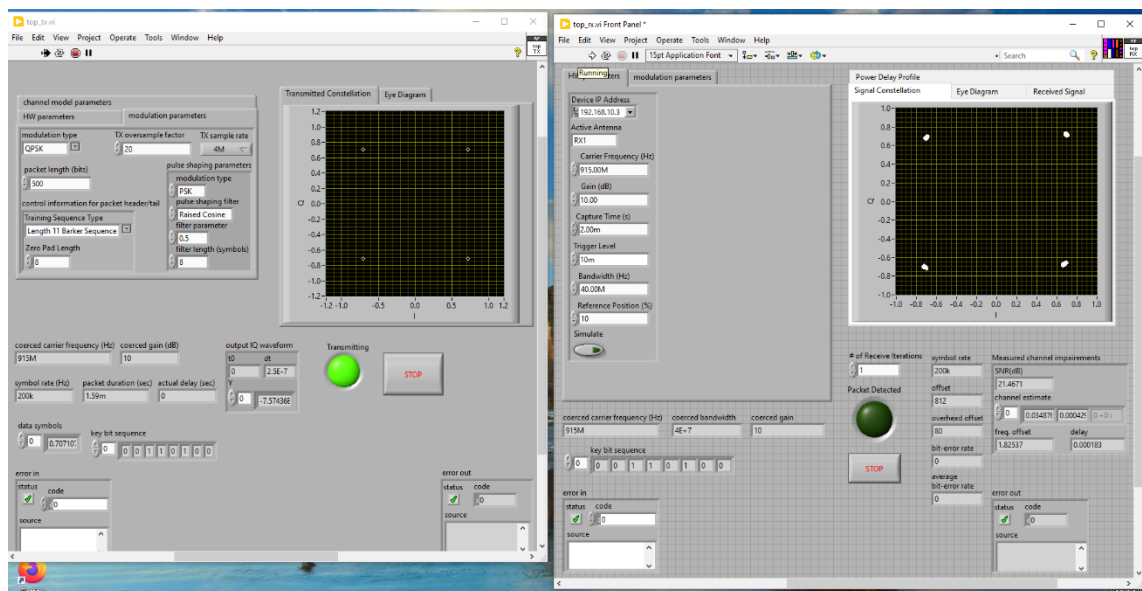


Figure 8: top\_tx.vi & top\_rx.vi (signal constellation)

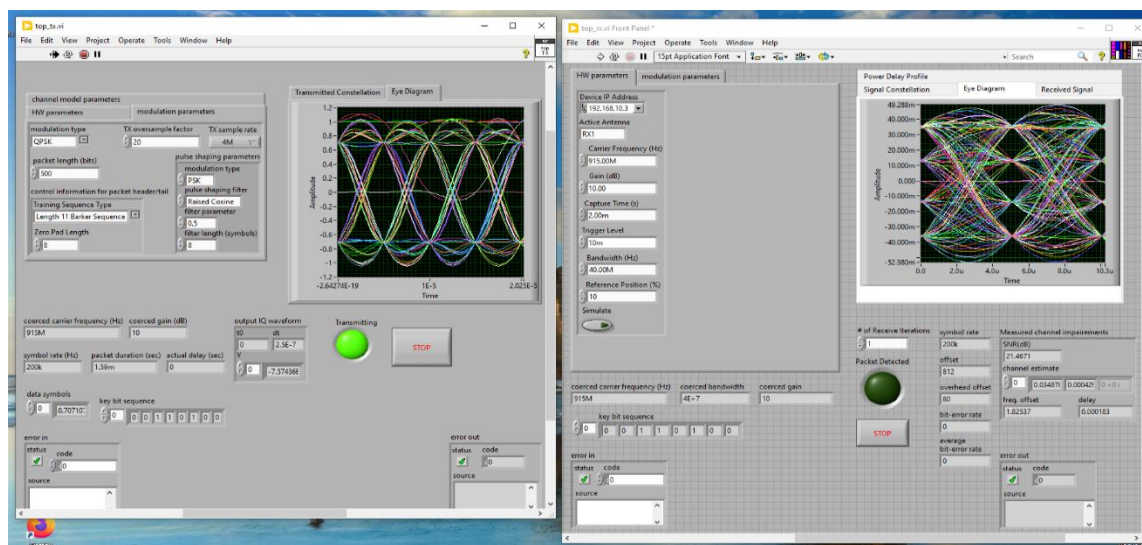


Figure 9: top\_tx.vi & top\_rx.vi (eye diagram)

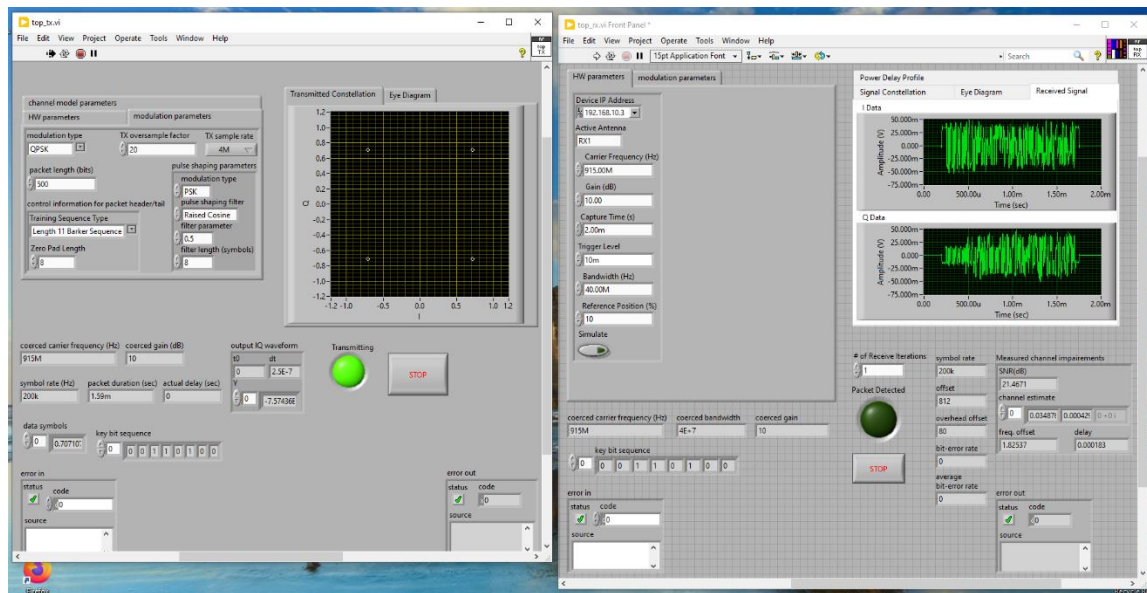


Figure 10: top\_tx.vi &amp; top\_rx.vi (received signal)

### Question and answers:

1. Before changing any of the settings, calculate an average value (using five runs or so) of the inherent offset between the transmitter and receiver.

**Ans:** 895.2 (895,896,895,895,895).

2. Based on the system parameters, what is the range of frequency offsets that can be estimated by your frequency offset estimation algorithm?

**Ans:** -500 KHz to 500 KHz

3. Let  $f_m$  be the maximum correctable frequency offset of your system (i.e., as calculated in the previous question). Modify the carrier frequency of your transmitter so that you will cause an effective offset of  $0.80f_m$  at the receiver. What is the new value of the carrier frequency at your transmitter?

**Ans:** carrier frequency = 915 MHz

from the formula new value of carrier frequency at transmitter will be  
 $915 \text{ MHz} + 0.80f_m = 915\text{M} + 0.80(500\text{k}) = 915\text{M} + 400\text{K} = 915.4\text{MHz}$ .

4. How does this frequency offset impact the performance of your correlation-based frame detector (i.e., does your correlation-based detector still find the start of the frame)? Describe what happens to the error rate and constellation of your system.

**Ans:** The correlation-based frame detector still finds the start of the frame at cost of high BER. And received constellation still distorted.



1. **Discuss any problems you might have encountered and how you overcame these obstacles.**

**Ans:** while implementing sliding correlator we have encountered problem with loops we overcome this with lab notes and proffser with his suggestions.

2. **n this lab you observed how your code operates over a real wireless link. Answer the following questions about your experience in lab.**

- a) **Name three impairments in a real wireless system which impact the accuracy of the correlation-based detection method.**

**Ans:** Frequency offset, Doppler Shift, Multipath fading